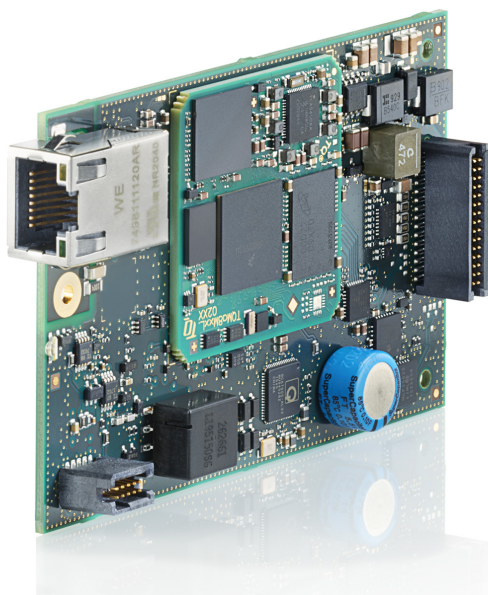


vSECC.single Board

Communication Board for Wallboxes and Single Charge Points

What is vSECC.single Board?

vSECC.single Board is a communication board for the fast development of smart AC and DC wallboxes and charging stations. The board can operate one charging connector and is designed to sit on top of customer electronics in the same housing. As part of the vSECC Controller family, the board comes with the vSECC Software. The controller handles the communication between the electric vehicle (EV), the charging station management system (CSMS) or Energy Management System (EMS), the power electronics (PE) for DC charging and other peripherals. vSECC.single Board communicates to the backend via the Open Charge Point Protocol (OCPP). Peripheral devices such as PE for DC charging can be controlled via the available hardware interfaces, e.g. CAN or Ethernet. The communication controllers support DC charging communication over Combined Charging System (CCS), North American Charging System (NACS) according to ISO 15118, DIN SPEC 70121 or AC charging communication for CCS Type 1 & 2 according to ISO 15118 and IEC 61851.



The vSECC.single board is the brain of any smart DC wallbox and easily integrated into customer electronics

Application Areas

The vSECC.single Board's compact design makes it perfect for smart charging stations or wallboxes, especially for bidirectional EV charging at home or work. In addition, the vSECC.single Board is also ideal for use in mobile chargers or any charging station serving a single charge point. This versatility allows photovoltaic system manufacturers to seamlessly integrate advanced charging communication capabilities into their products, enabling the efficient use of solar-generated electricity for charging electric vehicles.

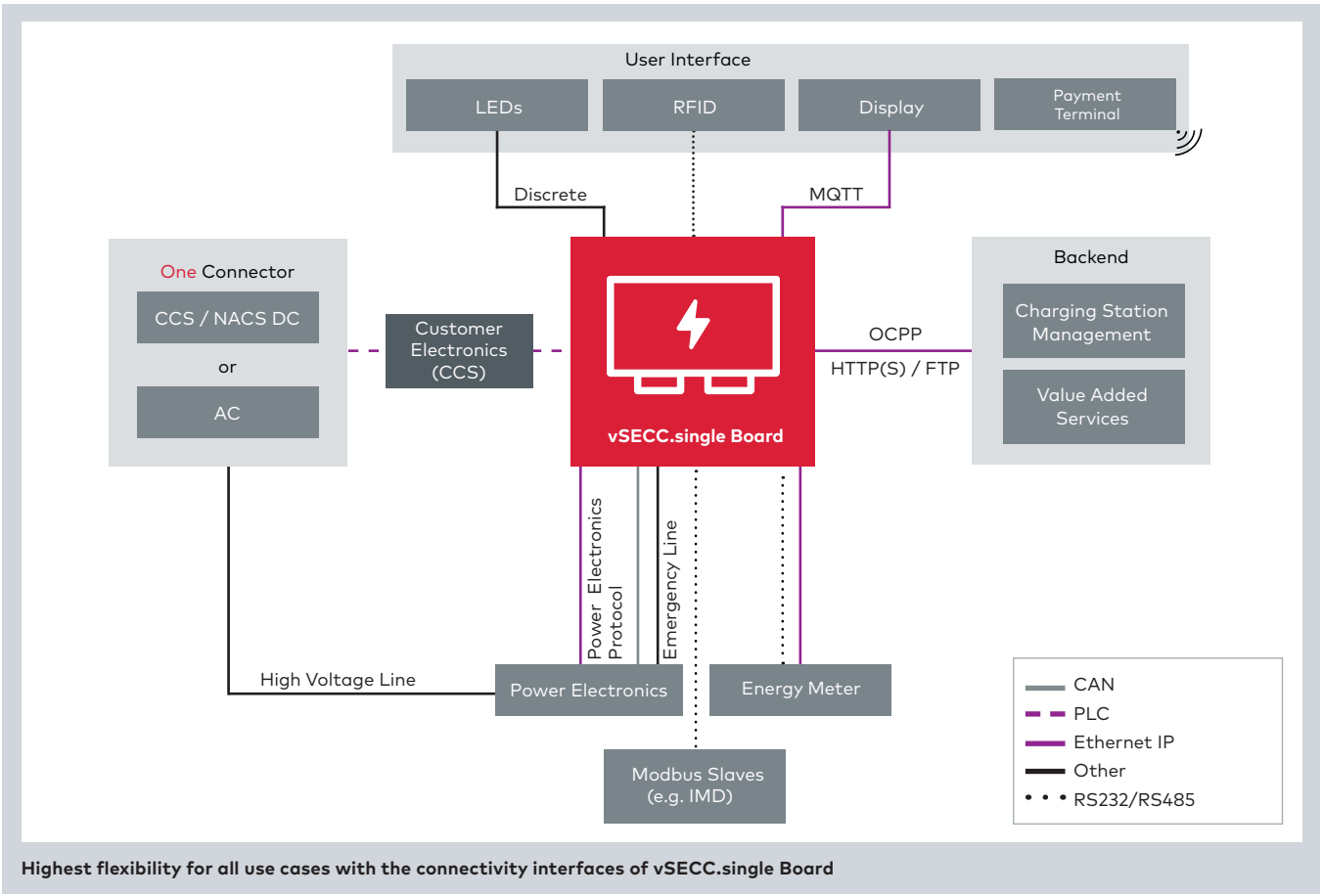
Overview of Advantages

- > Accelerates the development and paves the way into the emerging smart AC and DC wallbox market
- > Manages the entire communication complexity within the wallbox
- > Central control unit that offers maximum flexibility in the selection of peripheral devices and easy operation
- > Small form factor ideal for wallboxes and single dispenser charging stations
- > High performant i.MX 8 processor
- > Predestined for microgrids with energy management integration and bidirectional DC charging
- > Ongoing free software updates

Functions

- > Communication board handling the complete charging process for one charge point
- > Charging communication according to ISO 15118-2, -20, -3, DIN SPEC 70121 (DC)
- > Supports OCPP 1.6J and 2.0.1 for the CSMS communication via Ethernet, ready for higher OCPP versions
- > Additional functionality can be developed in the vSECC software
- > Supports DC Bidirectional Power Transfer (BPT)
- > Vehicle identification possible via Plug & Charge, External Identification Means (EIM), RFID and Autocharge
- > MQTT broker provides information e.g. about current charging process

More information: www.vector.com/vsecc



Technical Data

Feature	vSECC.single Board
Main CPU	i.MX8M Nano
Power supply	12 V DC
Communication	1 x CAN 1 x 1000Base-T / 100Base-TX
	1 x RS232 1 x RS485
Charging interfaces	IEC 61851-1 / -23 SAE J1772 DIN SPEC 70121 ISO 15118-2 / -3 ISO 15118-20
Inputs and outputs	10 x digital I/O (logic level 3.3 V) 4 x 0-5 V analog input
Temperature inputs	2 x temperature sensor input (PT1000)
Safety output	1 x (logic level 3.3 V)
Dimensions (WxHxD)	14.25 mm x 58 mm x 94.1 mm
Weight	~ 48 g
Technical usage and storage	Temperature: -40 °C to +70 °C
	Altitude: 0 – 2000m above sea level
Connectors	1 x RJ45 1 x USB (in development)
	1 x 30 pole SAMTEC base board connector 1 x 10 pole SAMTEC charging connector